

Chapter 9: Conclusions

Machine Translation: Advanced Topics

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- This course explored **modern machine translation**
- From classical approaches to large language models

Core question

How do machines translate — and what determines translation quality?

A historical trajectory

- Rule-based MT
- Statistical MT (SMT)
- Neural MT (encoder–decoder)
- Transformer models
- Large language models (LLMs)

Trend

From explicit rules → data-driven learning → general-purpose models

What changed?

- More data
- Larger models
- Better architectures
- Improved training methods

But

The training objective remains central

Two paradigms for translation

- **Translation-specialised models**

- encoder–decoder
- trained on parallel data
- explicit mapping

- **General-purpose language models**

- decoder-only
- trained on text
- translation emerges from prompting

Mapping vs. continuation

- Encoder–decoder:
 - source $\rightarrow$ target mapping
- Decoder-only:
 - text continuation
 - translation is emergent

Key distinction

How the model is trained determines how it translates

A fundamental trade-off

- **General-purpose LLMs**

- flexible
- fluent
- broad capabilities
- weaker adequacy/control

- **Translation models**

- reliable
- consistent
- strong adequacy
- less flexible

How to evaluate MT systems

- Always compare against baselines
- Use multiple metrics:
 - BLEU
 - chrF
 - newer metrics
- Inspect outputs manually

Important

Metrics do not capture everything

Reproducibility matters

- Results depend on:
 - data splits
 - preprocessing
 - decoding settings
 - randomness
- Report:
 - settings
 - averages
 - variance

What you learned

- Prepare and clean data
- Train and evaluate MT systems
- Use pretrained models
- Work with LLM prompting
- Use APIs for translation

LLMs in machine translation

- Strong fluency
- Easy to use (prompting, APIs)
- Useful for:
 - prototyping
 - low-resource scenarios

Limitation

No guaranteed adequacy or alignment

Central insight

- Translation quality is not only about model size
- It depends on:
 - objective
 - data
 - architecture

Take-away

Bigger models are better — but not fundamentally different

- MT is part of a larger NLP ecosystem
- Closely related to:
 - summarisation
 - dialogue systems
 - information extraction

Open challenges

- Faithfulness vs. fluency
- Low-resource languages
- Bias and fairness
- Evaluation limitations
- Control and reliability of LLMs

Future directions

- Better multilingual models
- Hybrid systems (LLM + MT)
- Improved evaluation
- More controllable generation

Final message

- There is no single best approach
- Each paradigm has strengths and weaknesses

Conclusion

Understanding the model is essential for using it effectively

Thank you

Questions?